

Harnessing bacteriophages to combat antibiotic-resistant chronic postoperative infections: A promising case series

Ajay Kumar, Alka Shukla, S. Alshad, Pankaj Kumar, Gopal Nath *

Virus Research and Diagnostic Laboratory (VRDL), Department of Microbiology, Institute of Medical Sciences, Banaras Hindu University, Varanasi, India

ARTICLE INFO

Keywords:

Bacteriophage therapy
Prosthetic joint infection
Multidrug-resistant
Phage isolation
Biofilm disruption

ABSTRACT

Background: Prosthetic joint infections (PJIs) caused by multidrug-resistant (MDR) bacteria and biofilm formation are often refractory to standard surgical and antibiotic care. Personalized bacteriophage therapy has emerged as a targeted alternative for such cases.

Methods: We present a descriptive case series of three patients with chronic MDR PJIs after total knee replacement (TKR) who were treated with locally applied high-titer lytic bacteriophage preparations. Clinical isolates were identified and characterized; environmental phages were isolated, purified, titrated, and tested for host specificity before local therapy. Outcomes were assessed clinically and microbiologically during follow-up.

Results: All three patients (MDR *Staphylococcus aureus*, extended-spectrum beta-lactamase-producing *Klebsiella pneumoniae*, and carbapenem-resistant *Pseudomonas aeruginosa*) achieved complete wound healing and resolution of infection while retaining their prostheses. Healing times were 56, 60, and 45 days, respectively; no recurrences were observed during the reported follow-up period. Phage titers ranged from 10^6 to 10^9 PFU/mL; therapy was well tolerated, with no reported adverse events.

Conclusion: In this series, personalized, locally applied bacteriophage therapy was as safe and effective as a prosthesis-preserving rescue strategy for refractory MDR PJIs. These findings support further clinical evaluation and protocol standardization for the use of phages in the treatment of complex orthopedic infections.

1. Introduction

Periprosthetic joint infection (PJI) is one of the complications with the highest severity and patient morbidity associated with arthroplasty. Patients may require extended hospital stays, undergo multiple additional surgical interventions, and PJI entails significant costs to the healthcare system [1,2]. PJI continues to be a persistent problem, even with the recent improvements made to surgical methods and the design of the implants [3]. A particular concern with PJI is the ability of certain infectious agents to create a biofilm on the surface of the implants. Bacteria within the biofilm undergo a phenotypic change to a sessile state, meaning they become less active, conduct differential gene expression, and show increased tolerance to chemotherapeutic products and the immune system of the host [4,5]. These infections are often chronic with the ability to relapse, and the only solution to completely remove the infection is complete removal of the implant.

The global spread of multidrug-resistant (MDR) organisms has increased the complexity of PJI management. Refractory PJIs are now routinely caused by pathogens such as methicillin-resistant

Staphylococcus aureus (MRSA) and *Klebsiella pneumoniae* that produce extended-spectrum beta-lactamases and *Pseudomonas aeruginosa* that are resistant to carbapenems, which significantly restricts the available antibiotic options [6,7]. In the context of established biofilms and MDR infections, standard treatment options - such as prolonged antimicrobial therapy, debridement with implant retention, one- or two-stage revision surgeries, and antimicrobial spacer placement - are associated with increased rates of failure [8,9]. As such, there is increasing demand for new or combination treatment strategies aimed at biofilm-related infections that conserve the infected prosthesis.

Bacteriophage therapy, a promising strategy against difficult-to-treat infections, including PJIs, has recently resurfaced [10]. Bacteriophage lysates are able to target infection-specific bacteria by biofilm disruption, infection site replication, and host-specific targeting [11,12]. Phages also work against bacteria that are resistant to antibiotics and can be tailored to the specific strain infecting a patient, making them especially valuable for infections that are multi-drug resistant (MDR) and where standard treatment options are no longer effective [13]. Recent small clinical studies and compassionate use reports have

* Corresponding author.

E-mail address: gopalnath@gmail.com (G. Nath).

<https://doi.org/10.1016/j.medmic.2026.100178>

Received 21 January 2026; Received in revised form 21 April 2026; Accepted 8 May 2026

Available online 21 May 2026

2590-0978/© 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

indicated that localized phages may help manage an infection, assist with wound healing, and help retain prosthesis in some cases of PJI [14, 15]. In this regard, the current report presents detailed microbiological assessment, treatment strategy, and clinical outcome for three cases of chronic multi-drug resistant (MDR) PJI using localized, patient-specific phage therapy.

2. Methods

2.1. Study design and setting

This study was structured as a descriptive case series involving three consecutive patients with chronic multidrug-resistant prosthetic joint infections (MDR PJIs). Patients with chronic MDR PJIs had non-healing wounds that did not heal despite exposure to conventional antibiotic treatment, which defied traditional abscess management. All included patients had previously undergone prosthetic joint implantation and were classified as having chronic infections based on clinical and/or microbiological criteria [16,17]. Only cases that provided informed consent were included. Patients were excluded if they had acute prosthetic joint infections, infections caused by organisms sensitive to standard antibiotic therapy, or an adequate response to conventional antibiotic treatment. Additionally, individuals with non-prosthetic joint infections, incomplete clinical data or follow-up, or those who declined to provide consent were not included in the study.

The patients were treated with personalized bacteriophage preparations at the Department of Microbiology, Institute of Medical Sciences (IMS), Banaras Hindu University (BHU), Varanasi. Ethical approval was obtained from the Institutional Ethics Committee (Dean/2022/EC/3329), and written informed consent was secured from all patients before the initiation of bacteriophage therapy.

2.2. Clinical evaluation and specimen collection

The clinical assessment included inspection of the wound, measurement of the wound area, quantification of the wound discharge, determination of the level of pain and inflammation, and a functional assessment. Specimens (pus aspirate, wound exudate, deep tissue swab) were collected under aseptic conditions and, when possible, after an antibiotic-free interval of 48 hours. Samples were collected and stored at 4 °C until processed.

2.3. Microbiology and antibiotic susceptibility

The antibiotics were discontinued because they proved ineffective in treating the wounds, and after 15 days off antibiotics, the specimens were collected. The specimens were cultured on nutrient, blood, and MacConkey agar, then Gram-stained and identified using standard biochemical methods. Antibiotic susceptibility testing was performed in accordance with CLSI 2025 guidelines using the Kirby-Bauer disk diffusion method to confirm MDR phenotypes [18] (Fig. 1A–C). Isolates were stored on appropriate media for phage screening.

2.4. Phage isolation, purification, and characterization

The patient isolate was used as a host to enrich environmental samples (local drains and sewers close to the hospital). Phage isolation was performed using double-layer agar (plaque) assays; the phages were plaque-purified, amplified, and then titrated to 10^6 – 10^9 PFU/mL. The range and specificity of the (lytic) phages were assessed and confirmed through spot tests on the panel of clinical isolate(s) [19]. For therapeutic use, phages exhibiting significant lytic activity and appropriate host range were selected (Fig. 1D–G). As part of quality control, endotoxicity and sterility tests were done. Phage preparations for clinical applications underwent sterility tests to detect the absence of bacterial and fungal contamination. Additionally, to ensure the phage preparations had

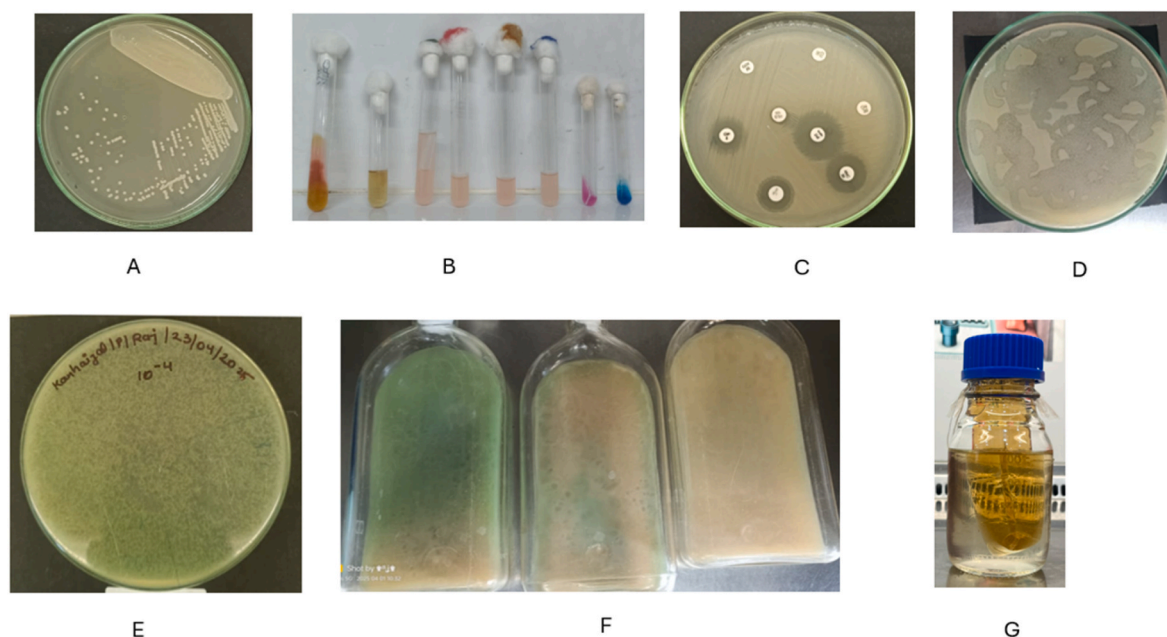


Fig. 1. The sequential workflow for bacteriophage isolation, characterization, and production. (A) The process begins with isolating the target bacterium, which is confirmed by observing well-defined bacterial colonies on culture media. (B) Biochemical tests are performed to identify and characterise the bacterial isolate based on metabolic reactions. (C) Antibiotic susceptibility testing (AST) is performed to determine the bacterium's resistance profile. (D) Bacteriophage spotting is then performed on a bacterial lawn to screen for lytic phages, indicated by clear zones of lysis. (E) Phages are further confirmed by the double-layer agar plaque assay, in which distinct plaques indicate phage infectivity and specificity. (F) Selected lytic phages are produced in bulk in Roux bottles to obtain high-titre phage lysate. (G) Finally, the phage lysate undergoes membrane dialysis to remove impurities and get a clear, concentrated bacteriophage preparation suitable for downstream therapeutic applications.

optimal (select) storage temperatures, their (the) enzyme activity (was) regulated to fall within the (normal) physiological range, and the samples were refrigerated at 4 °C before use. The pH of the preparations was (also maintained to) within the (normal) physiological range (to preserve) phage activity and ensure (the) optimal storage temperatures (to preserve) phage (to preserve) stability.

The endotoxin level of the harvested phage preparations was determined using the “Thermo Scientific™ Pierce™ LAL Chromogenic Endotoxin Quant” ELISA-based kit, according to the manufacturer's instructions. The endotoxin level was within the safe range of 0.005–0.01 EU/mL [20]; therefore, the phages were considered free of endotoxin.

2.5. Treatment protocol and supportive care

The bacteriophage therapy was applied directly to the infected skin area, ensuring optimal contact with the bacterial target. Each time this treatment was applied, the wound was first thoroughly cleaned with sterile normal saline (0.9% NaCl) to remove any debris, pus, or exudate; extra care was taken to protect the bacteriophages from deactivation by any antiseptic agents. After cleansing, the wound was allowed to air-dry for approximately 10 minutes to minimize dilution of the liquid bacteriophage preparation to be used. A volume of 0.5 to 1.0 mL of the liquid bacteriophage preparation was applied directly to the wound surface using a sterile syringe, and the solution was dropped onto and over the wound without applying any pressure to prevent tissue disruption and ensure an even distribution. A sterile gauze dressing saturated with the same liquid bacteriophage preparation was placed over the wound to maintain prolonged contact between the bacteriophage and their target bacteria. Once the gauze dressing had been placed in position, the bacteriophage-soaked dressing was left in place for approximately 20 min; then a separate sterile outer dressing was placed over the bacteriophage-soaked dressing, and both were left undisturbed for at least 8 h. The frequency of phage application was tailored and adjusted daily based on wound characteristics, including appearance, exudate volume, signs of local inflammation, and overall clinical response. In this study, bacteriophage therapy was administered once daily. As clinical improvement became evident, characterized by reduced discharge, formation of healthy granulation tissue, and a progressive reduction in wound size, the treatment was continued accordingly. Wound progress was monitored daily by recording wound size, granulation tissue formation, reduction in pus or odour, and patient-reported pain.

Importantly, no antibiotic therapy was given during the entire course of bacteriophage treatment, enabling uninterrupted phage-bacterium interaction and independent evaluation of the phage-therapeutic effect.

2.6. Outcome measures

The primary outcomes consisted of clinical wound healing, defined as wound closure; resolution of wound discharge, defined as the absence of purulent or serous drainage; and sterilization of follow-up wound cultures after completion of bacteriophage therapy to demonstrate microbiological clearance.

Secondary outcomes included the time to wound healing (measured in days from the initiation of phage therapy to complete closure), retention of the prosthetic joint, any recurrence of infection during the follow-up period, and any adverse events or complications related to phage therapy.

Follow-up assessments included multiple clinical evaluations (wound inspections, measurements of wound size, assessments of local inflammation, and documentation of functional improvements), microbiological assessments of the wound, and repeated cultures to verify that the previously identified pathogens were no longer present.

3. Case presentations

3.1. Case 1 - *Staphylococcus aureus* (MRSA)-penicillin, methicillin, and cephalosporins resistant

A 66-year-old woman with a history of right total knee replacement presented with a chronic prosthetic joint infection marked by persistent seropurulent discharge and a non-healing surgical wound. Despite undergoing multiple surgical debridements, vacuum-assisted closure (VAC) therapy, placement of an antibiotic spacer, and subsequent revision total knee arthroplasty, adequate wound healing was not achieved, and the patient developed a persistent draining sinus tract, indicating a chronic and treatment-resistant infection. Due to the failure of standard surgical and antimicrobial approaches, the patient was referred for bacteriophage therapy as a salvage option.

Microbiological analysis of wound cultures consistently identified methicillin-resistant *Staphylococcus aureus* (MRSA), confirming the presence of a multidrug-resistant pathogen. A personalized bacteriophage preparation comprising three lytic staphylococcal phages (S1–S3), with titers ranging from 10^6 to 10^8 PFU/mL, was formulated and administered locally daily, alongside ongoing supportive wound care and close clinical monitoring.

Following the initiation of phage therapy, the patient demonstrated gradual but sustained clinical improvement, characterized by a progressive reduction in wound discharge, resolution of local inflammation, and a steady decrease in wound size (Figs. 2A and 3A). Complete wound healing was achieved by day 56, with restoration of healthy epithelial tissue. Subsequent follow-up wound cultures were sterile, the prosthetic joint was successfully retained, and no recurrence of infection was observed during the follow-up period.

3.2. Case 2 - *Klebsiella pneumoniae* (ESBL)penicillins, cephalosporins, and aztreonam-resistant

A 50-year-old female with a history of left total knee replacement presented with a recurrent prosthetic joint infection following revision surgery, which included placement of an antibiotic spacer and split-thickness skin grafting. Despite multiple surgical interventions and prolonged antimicrobial therapy, the patient developed a persistent draining sinus tract associated with surrounding tissue necrosis and poor wound healing, indicating a chronic and refractory infection. In view of the failure of conventional management strategies, the patient was referred for bacteriophage therapy.

Microbiological analysis of wound specimens consistently identified an extended-spectrum beta-lactamase (ESBL)producing *Klebsiella pneumoniae*. A personalized bacteriophage cocktail comprising four lytic *K. pneumoniae* phages (KP1–KP4), with titers ranging from 10^6 to 10^8 PFU/mL, was prepared and administered locally daily, in conjunction with standard wound care protocols and surgical debridement as required.

Following the initiation of phage therapy, the wound demonstrated gradual and sustained clinical improvement, characterized by reduced discharge, resolution of necrotic tissue, and progressive granulation. Complete wound closure was achieved by day 60 (Fig. 2: B and 3: B). At subsequent follow-up, the patient had regained full knee joint range of motion, the prosthesis was successfully retained, and no recurrence of infection was observed.

3.3. Case 3 - *Pseudomonas aeruginosa* (carbapenem-resistant)

A 55-year-old woman with a known history of rheumatoid arthritis on disease-modifying antirheumatic drugs (DMARDs) presented with an early postoperative infection after right total knee replacement. Over time, the condition progressed into a chronic ulcerative wound approximately 12 cm in size, with persistent purulent discharge and impaired wound healing. Despite extended, repeated courses of broad-



(caption on next page)

Fig. 2. Images depicting wound healing progression following personalized bacteriophage therapy. A) Case 1: Serial clinical photographs showing progressive healing of the wound at different time points during bacteriophage therapy. B) Case 2: Sequential clinical images documenting the wound condition at multiple time points during bacteriophage therapy, showing gradual reduction in wound area, improved granulation tissue formation, and Complete wound healing. and C) Case 3: Serial clinical photographs showing progressive wound healing at different stages during bacteriophage therapy, with visible reduction in wound length, decreased inflammation, and gradual restoration of normal skin architecture.

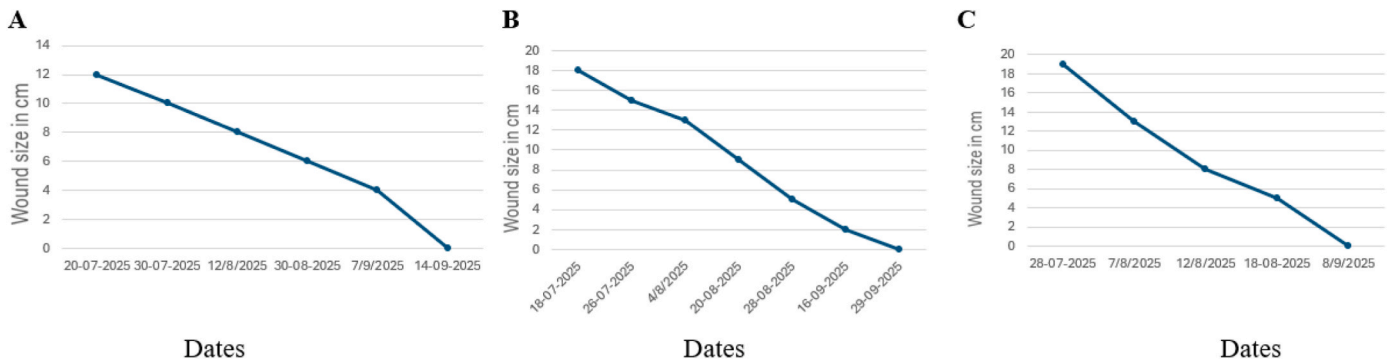


Fig. 3. Graphical representation of wound size reduction (in cm) during the course of bacteriophage treatment, demonstrating a continuous and significant decrease in wound dimensions over time. A) Case 1, B) Case 2 and C) Case 3.

spectrum antibiotics, the infection remained resistant to standard treatment, prompting further investigation into bacteriophage therapy. Microbiological examination of wound cultures consistently identified carbapenem-resistant *Pseudomonas aeruginosa*, indicating a multidrug-resistant pathogen with limited therapeutic options. A personalized bacteriophage cocktail comprising four lytic phages.

Aeruginosa phages (PS1–PS4), with titers ranging from 10^6 to 10^9 PFU/mL, were formulated and administered locally daily alongside standard wound care measures. Following the start of phage therapy, the patient showed rapid and sustained clinical improvement, characterized by a significant reduction in wound discharge, resolution of local inflammation, and progressive formation of granulation tissue. The wound size decreased markedly, and complete healing was achieved at the final follow-up, approximately 45 days later (Figs. 2C and 3C). The prosthetic joint was successfully retained, and the patient remained clinically stable and free of infection during follow-up.

4. Results

Personalized local bacteriophage therapy successfully controlled infections and promoted wound healing in all three patients with multidrug-resistant prosthetic joint infections. High-titre, host-specific lytic phages were applied directly to the site and were well tolerated, with no side effects. In all cases, there was a significant reduction in wound discharge, inflammation, and size, leading to complete healing. Notably, all prostheses were preserved; follow-up cultures remained negative; and no infection reoccurred, demonstrating long-term eradication and full clinical recovery. All findings are summarised in Table 1.

In cases involving *Staphylococcus aureus*, persistent postoperative discharge and wound dehiscence fully resolved, with complete epithelialization by Day 56 and negative follow-up cultures. The *Klebsiella pneumoniae* infection also resulted in complete wound healing by Day 60, restoring full knee range of motion and showing no recurrence. For *Pseudomonas aeruginosa*, an initial 12.0 cm ulcer with purulent discharge shrank to 4.0 cm within 20 days and healed completely by the final visit.

Overall, these results highlight the success of personalized bacteriophage therapy in eliminating infections, facilitating wound healing, and protecting prosthetic joints in resistant MDR infections.

5. Discussion

The present case series adds to the scarce and evolving body of evidence assessing the clinical utility of personalized bacteriophage therapy in multidrug-resistant prosthetic joint infections. Given the marked paucity of well-designed studies in this domain, our findings provide preliminary support for its potential role as an adjunctive or salvage therapeutic strategy for refractory periprosthetic joint infections caused by MDR pathogens. PJIs represent a therapeutic challenge due to the convergence of biofilm formation, limited antibiotic penetration, and increasing antimicrobial resistance, often necessitating complex revision surgeries or implant removal. In this context, achieving infection control while retaining the prosthesis remains a key yet challenging goal. The outcomes observed across the three cases described here are consistent with emerging clinical experiences indicating that locally applied, strain-specific lytic phages may help overcome some of these barriers when conventional approaches have failed [15,21].

A major conceptual advantage of bacteriophage therapy in PJI lies in its biological specificity and mechanism of action. Unlike antibiotics, lytic phages selectively infect and replicate within their bacterial hosts, enabling localized amplification at the site of infection and minimizing off-target effects on the commensal microbiota [11,12]. The proposed mechanism involves bacteriophage-mediated penetration of biofilms through a combination of passive diffusion and enzymatic degradation of the extracellular polymeric matrix. Specifically, many bacteriophages encode depolymerases—specialized enzymes that target and degrade polysaccharide components such as capsular polysaccharides, exopolysaccharides, and lipopolysaccharides [22]. By dismantling these structural barriers, depolymerases facilitate phage access to bacterial surface receptors, enabling infection of metabolically active cells within the biofilm and subsequent local amplification of antibacterial activity.

Table 1
Clinical characteristics and outcomes of three cases treated with bacteriophage therapy for multidrug-resistant prosthetic infections.

Case	Pathogen	Bacteriophages	Peak titer (PFU/mL)	Healing time (days)	Prosthesis retained	Infection Recurrence
1	<i>S. aureus</i> (MRSA)	S1–S3	10^6 – 10^8	56	Yes	No
2	<i>K. pneumoniae</i> (ESBL)	KP1–KP4	10^6 – 10^8	60	Yes	No
3	<i>P. aeruginosa</i> (CR)	PS1–PS4	10^6 – 10^9	45	Yes	No

Mechanistically, these enzymes function as hydrolases or lyases, cleaving glycosidic bonds via distinct biochemical pathways. Typically associated with phage tail fibers or spikes, depolymerases promote biofilm disruption by weakening matrix integrity and generating channels that permit deeper phage infiltration. Although not directly bacteriolytic, their activity enhances phage propagation and improves the penetration and efficacy of co-administered antibiotics and host immune factors. Collectively, these properties underscore their potential as adjunctive agents in the management of biofilm-associated infections, particularly in refractory settings such as prosthetic joint infections [12, 13]. Although biofilm disruption was not directly quantified in this study, the observed clinical trajectories are in line with prior reports suggesting that phage-mediated biofilm targeting may facilitate wound resolution and microbial clearance in orthopaedic infections [15,18].

An important practical insight from this series is the feasibility of sourcing effective therapeutic phages from local environmental reservoirs through targeted enrichment using the patient's clinical isolate. This approach enabled the rapid identification of lytic phages active against diverse MDR pathogens, including MRSA, ESBL-producing *Klebsiella pneumoniae*, and carbapenem-resistant *Pseudomonas aeruginosa*. While this strategy is particularly relevant in resource-limited settings, its broader clinical translation requires careful attention to quality control parameters, including sterility assurance, endotoxin burden, and reproducibility of phage preparations [11,14]. Standardization of phage manufacturing under regulatory frameworks remains a critical bottleneck for wider implementation and underscores the need for centralized or accredited phage production facilities.

The potential for bacterial resistance to phages represents another important consideration. Similar to antibiotics, bacteria can also develop resistance against phages. Bacterial resistance to bacteriophages is mediated through multiple defence mechanisms operating at both the surface and intracellular levels. Surface-associated defences, including receptor modification, capsular polysaccharide production, and biofilm formation, impede phage adsorption and limit access to host cell receptors. Intracellular defence systems, such as restriction–modification (R–M) systems and abortive infection (Abi) pathways, function by degrading invading phage nucleic acids or inducing programmed cell death to prevent phage replication and dissemination. In addition, adaptive immune mechanisms, most notably CRISPR–Cas systems, though not common or primary mechanisms, enable sequence-specific recognition and targeted neutralization of previously encountered phages, thereby conferring acquired resistance [23]. Clinical and experimental studies have demonstrated that such resistance can emerge in vivo, particularly when single-phage preparations are used [24]. In this series, the use of phage cocktails targeting multiple bacterial receptors was intended to mitigate this risk. Additional strategies, including adaptive phage training, sequential phage replacement, and rational phage-antibiotic combinations, phage-blue light combinations, ultrasound and phage are increasingly being explored to address resistance dynamics [15,21,25–27]. Incorporation of whole-genome sequencing and longitudinal molecular monitoring could further enhance understanding of host-phage interactions and resistance evolution during therapy, as demonstrated in recent clinical case analyses [15].

It is also important to emphasize that phage therapy should not be viewed as a standalone intervention for PJI. Surgical debridement, wound care optimization, and careful patient selection remain foundational components of management. In the cases presented here, phage therapy was integrated into a broader multidisciplinary treatment strategy rather than replacing standard care. This complementary role aligns with current expert opinion and published clinical experience, which advocate for phages as adjuncts rather than substitutes for established surgical and antimicrobial approaches [15,18,19].

6. Limitations

Several limitations of this study should be acknowledged. The small sample size and non-randomized, uncontrolled case-series design limit generalizability and preclude robust causal inference, with a potential for selection bias. Microbiological follow-up was primarily culture-based, which may underestimate low-level bacterial persistence within biofilms; incorporating molecular approaches, such as quantitative PCR or metagenomic sequencing, could provide a more sensitive assessment of treatment response. Additionally, phage–antibiotic interactions and the development of phage resistance were not systematically evaluated, and standardized pharmacokinetic or pharmacodynamic analyses were not feasible. Although no adverse effects were observed during short-term follow-up, the study's recent nature limits long-term data, preventing definitive conclusions regarding sustained safety and efficacy. Furthermore, host immune responses were not assessed. Despite these constraints, such case series offer valuable preliminary insights into feasibility and safety, and support hypothesis generation in this emerging field.

7. Conclusion

In summary, this case series underscores the translational relevance of personalized bacteriophage therapy in the management of complex periprosthetic joint infections involving multidrug-resistant organisms. The findings support the feasibility of integrating strain-specific phage preparations into existing clinical workflows for refractory orthopaedic infections. While definitive conclusions regarding efficacy require controlled studies, these observations reinforce the need for continued clinical evaluation, standardized regulatory frameworks, and collaborative efforts to advance phage therapy from compassionate use toward broader clinical adoption. As antimicrobial resistance continues to limit conventional treatment options, bacteriophages may emerge as an important component of future infection management strategies in orthopaedics.

Ethics statement & patient consent

Ethical approval was obtained from the Institutional Ethics Committee, Institute of Medical Sciences (IMS), Banaras Hindu University, Varanasi (approval number: **Dean/2022/EC/3329**). Written informed consent for treatment and for the publication of anonymized clinical details was obtained from all patients prior to therapy.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT to correct grammar and refine the manuscript's language. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Funding

No funding received.

CRediT authorship contribution statement

Ajay Kumar: Data curation, Investigation, Methodology. **Alka Shukla:** Data curation, Writing – original draft, Writing – review & editing. **S. Alshad:** Methodology, Writing – original draft. **Pankaj Kumar:** Investigation, Methodology. **Gopal Nath:** Conceptualization, Supervision, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors gratefully acknowledge the participation and cooperation of the patients involved in this study. The authors also acknowledge support from the Department of Health Research, Ministry of Health & Family Welfare, Government of India, New Delhi, and the Indian Council of Medical Research (ICMR) for the establishment of the State-Level Virus Research and Diagnostic Laboratory (VRDL). Additionally, the authors acknowledge the use of AI-assisted tools for language editing and grammatical refinement of the manuscript.

References

- [1] Tande AJ, Patel R. Prosthetic joint infection. *Clin Microbiol Rev* 2014;27(2):302–45.
- [2] Kurtz SM, Lau E, Schmier J, Ong KL, Zhao K, Parvizi J. Infection burden for hip and knee arthroplasty in the United States. *J Arthroplast* 2008;23(7):984–91.
- [3] Parvizi J, Gehrke T. Definition of periprosthetic joint infection. *J Arthroplast* 2014;29(7):1331.
- [4] Costerton JW, Stewart PS, Greenberg EP. Bacterial biofilms: a common cause of persistent infections. *Science* 1999;284(5418):1318–22.
- [5] Donlan RM, Costerton JW. Biofilms: survival mechanisms of clinically relevant microorganisms. *Clin Microbiol Rev* 2002;15(2):167–93.
- [6] Otto M. MRSA virulence and spread. *Cell Microbiol* 2012;14(10):1513–21.
- [7] Bonomo RA, Burd EM, Conly J, et al. Carbapenemase-producing organisms: a global scourge. *Clin Infect Dis* 2018;66(8):1290–7.
- [8] Zimmerli W, Trampuz A, Ochsner PE. Prosthetic-joint infections. *N Engl J Med* 2004;351(16):1645–54. paperpaper.
- [9] Osmon DR, Berbari EF, Berendt AR, et al. Diagnosis and management of prosthetic joint infection. *Clin Infect Dis* 2013;56(1):e1–25.
- [10] Abedon ST, García P, Mullany P, Aminov R. Phage therapy: past, present and future. *Front Microbiol* 2017;8:981.
- [11] Harper DR, Parracho HMRT, Walker J, et al. Bacteriophages and biofilms. *Antibiotics* 2014;3(3):270–84.
- [12] Sillankorva S, Oliveira R, Azeredo J. Bacteriophages and their role in biofilm control. *Crit Rev Microbiol* 2012;38(4):349–70.
- [13] Chan BK, Abedon ST. Phage therapy pharmacology. *Pharmaceutics* 2015;7(4):491–526.
- [14] Ferry T, Kolenda C, Batailler C, et al. Phage therapy as adjuvant to conservative surgery and antibiotics for prosthetic joint infection. *Clin Infect Dis* 2020;71(10):e702–5.
- [15] Aslam S, Pretorius V, Lehman SM, et al. Novel bacteriophage therapy for multidrug-resistant prosthetic joint infection. *Open Forum Infect Dis* 2019;6(3). ofz005.
- [16] McNally M, Sousa R, Wouthuyzen-Bakker M, Chen AF, Soriano A, Vogely HC, Clauss M, Higuera CA, Trebse R. The EBJIS definition of periprosthetic joint infection: a practical guide for clinicians. *Bone Jt J* 2021;103-B:18–25.
- [17] Youssef Y, Roschke E, Dietze N, Dahse AJ, Chaberny IF, Ranft D, Pempe C, Goralski S, Ghanem M, Kluge R, Lübbert C, Rodloff AC, Roth A. Early-outcome differences between acute and chronic periprosthetic joint Infections-A retrospective single-center study. *Antibiotics (Basel)* 2024 Feb 20;13(3):198.
- [18] Schuetz AN, Ferrell A, Hindler JA, Humphries R, Bobenchik AM. Overview of changes in the clinical and laboratory standards institute performance standards for antimicrobial susceptibility testing: M100 32nd and 33rd editions. *J Clin Microbiol* 2025;63(9):e0162323.
- [19] Shende RK, Hirpurkar SD, Sannat C, Rawat N, Pandey V. Isolation and characterization of bacteriophages with lytic activity against common bacterial pathogens. *Vet World* 2017;10(8):973–8.
- [20] Gorbet MB, Sefton MV. Biomaterial-associated thrombosis: roles of coagulation factors, complement, platelets and leukocytes. *Biomaterials: Silver Jubilee Compendium* 2004 Jan 1:219–41.
- [21] Dedrick RM, Guerrero-Bustamante CA, Garlena RA, et al. Engineered bacteriophages for treatment of a patient with a disseminated drug-resistant *Mycobacterium abscessus* infection. *Nat Med* 2019;25(5):730–3.
- [22] Knecht LE, Veljkovic M, Fieseler L. Diversity and function of phage encoded depolymerases. *Front Microbiol* 2020 Jan 10;10:2949.
- [23] Ibrahim R, Aranjani JM. Bacterial defense mechanisms against bacteriophages: an evolutionary arms race. *Arch Microbiol* 2026 May;208(5):229.
- [24] Pirnay JP, De Vos D, Verbeken G, et al. The phage therapy paradigm: prêt-à-porter or sur-mesure? *Pharm Res* 2011;28(4):934–93.
- [25] Shukla A, Seyyadali A, Rastogi S, Rani S, Nath G. Synergistic potential of bacteriophage and blue light therapy against biofilm-associated *Klebsiella pneumoniae* in postoperative gynaecological infections. *Photodiagnosis Photodyn Ther* 2026 Feb 18:105399.
- [26] Yuan L, Ma L, Liu Y, Jiang Y, Xiong D, Xu Z, Yang Z, Jiao XA. Synergistic combination of ultrasound and bacteriophages therapy for effective control of biofilm formation by multidrug-resistant *Salmonella typhimurium*. *Innov Food Sci Emerg Technol* 2025 Jul 18:104117.
- [27] Malik S, Nehra K, Rana JS. Bacteriophage cocktail and phage antibiotic synergism as promising alternatives to conventional antibiotics for the control of multi-drug-resistant uropathogenic *Escherichia coli*. *Virus Res* 2021 Sep 1;302:198496.